ACE Semantic Gateway

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Code repository:

https://github.com/rosatisoft /ace-minimum-energy-criterion

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Abstract

Large language models (LLMs) frequently generate outputs that appear coherent yet are not grounded in a stable interpretation of the input. These hallucinations are commonly attributed to limitations of probabilistic language modeling, but we argue that they also arise from a structural property of current inference architectures: the semantic reference frame used during generation is typically derived implicitly from the prompt itself.

When prompts are ambiguous or contextually underspecified, multiple semantic interpretations may coexist within the model’s embedding space. Generation may then

proceed within an unstable prompt-derived subspace, producing responses that are locally coherent but globally misaligned with the intended context.

This paper introduces **Contextual Reference Fields**, a framework that separates context identification from response generation. Instead of deriving the semantic frame directly from the prompt, the system first identifies the most plausible contextual domain using a contextual matrix and constructs a structured reference field composed of domain anchors, axioms, and knowledge elements. Candidate inputs are then evaluated relative to this field using the **ACE origin energy criterion**, which measures semantic deviation from the contextual subspace.

We implement this approach as a prototype middleware layer, the **ACE Semantic Gateway**, which performs contextual classification, constructs contextual reference fields, and evaluates semantic alignment prior to generation. This architecture provides a deterministic mechanism for mitigating semantic instability and suggests an alternative design paradigm for language model inference.

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1. Introduction

Large language models (LLMs) based on transformer architectures have achieved remarkable success across a wide range of natural language tasks. These systems are capable of generating coherent text, answering complex questions, and performing reasoning tasks across multiple domains. Despite these advances, a persistent limitation remains: LLMs frequently produce outputs that appear plausible but are not grounded in stable semantic interpretation. This phenomenon is commonly referred to as **hallucination**.

Hallucinations occur when a model generates information that is syntactically coherent yet semantically unsupported by the input or by reliable knowledge sources. While several mitigation strategies have been proposed—including retrieval augmentation, reinforcement learning from human feedback, and post-generation filtering—these approaches typically address hallucination after the generation process has already begun.

This work explores the hypothesis that hallucinations are not merely a byproduct of probabilistic language modeling, but are partly caused by a **structural instability in the semantic reference frame used during inference**.

In most current architectures, the semantic frame used to interpret an input is implicitly derived from the prompt itself. Because natural language prompts may be incomplete, ambiguous, or contextually under-specified, the model must infer the appropriate semantic domain probabilistically. When multiple interpretations are plausible, the model may construct competing semantic frames within its internal representation space.

As a result, generation may proceed within a semantic environment that does not correspond to the intended meaning of the input. The produced response may therefore appear coherent while being grounded in an incorrect contextual interpretation.

We describe this condition as **prompt-derived semantic instability**.

This paper introduces a different architectural approach to semantic interpretation in language model systems. Instead of deriving the semantic frame directly from the prompt, we propose the use of **Contextual Reference Fields**, constructed from a structured contextual matrix that precedes generation.

In this architecture, the system first performs contextual analysis of the input in order to identify the most appropriate semantic domain. From this domain, a contextual reference field is constructed using domain-specific anchors, axioms, and knowledge elements. These elements define a stable semantic environment from which a reference subspace can be derived.

The input representation is then evaluated relative to this contextual subspace using the **ACE (Axiomatic Criterion Engine) origin energy criterion**, which measures the deviation of the input from the contextual field.

By separating **context identification** from **response generation**, the proposed architecture reduces the likelihood that the model will generate responses within unstable prompt-derived semantic spaces.

To demonstrate this approach, we implement a prototype middleware system referred to as the **ACE Semantic Gateway**, which operates as a deterministic semantic evaluation layer placed before the language model. The gateway performs contextual classification, constructs a contextual reference field, evaluates alignment using the origin energy criterion, and determines whether generation should proceed.

The contributions of this work are threefold:

1. We identify a structural limitation in prompt-derived inference that contributes to hallucination in language models.
2. We introduce the concept of **Contextual Reference Fields**, which provide a stable semantic environment for evaluating input alignment prior to generation.
3. We present a practical middleware implementation, the **ACE Semantic Gateway**, demonstrating how contextual reference fields and origin energy evaluation can be integrated into language model systems.

Together, these contributions suggest that improving semantic stability in language model inference may require architectural changes that separate contextual interpretation from generative processes.

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2. Related Work

Research related to the present work can be organized into four overlapping areas: transformer architectures, hallucination in large language models, uncertainty-based detection, and retrieval or guardrail-based mitigation.

2.1 Retrieval-Augmented and External Grounding Approaches

Another major mitigation strategy is retrieval-augmented generation (RAG), in which models consult external knowledge sources during inference. The original RAG framework demonstrated that combining parametric generation with retrieved evidence improves factuality and task performance in knowledge-intensive settings.

This approach is highly relevant because it externalizes part of the model's knowledge. However, retrieval alone does not fully solve the problem addressed in this paper. Retrieval can improve factual grounding, but the semantic frame in which retrieved information is interpreted may still be unstable if the context of the request is not properly identified first. Our approach is therefore complementary: contextual reference fields can serve as a semantic stabilization layer before or alongside retrieval-based generation.

2.2 Prompt Injection and Prompt-Level Guardrails

A related body of work studies prompt injection and instruction manipulation. These attacks exploit the fact that prompt-following systems often treat input text simultaneously as content and control signal, enabling malicious or misleading instructions to alter system behavior. Prior studies have shown that prompt injection is a practical vulnerability in LLM-integrated applications and that the model's instruction-following flexibility can itself become an attack surface.

This literature reinforces one of the main claims of the present work: systems that rely exclusively on prompt-derived control remain structurally fragile. The ACE Semantic Gateway differs from prompt-only guardrails by moving part of semantic control into a deterministic contextual layer, where identity of domain, contextual alignment, and semantic deviation can be evaluated prior to full generative inference.

2.3 Position of the Present Work

Taken together, prior work shows that hallucination can be studied through factuality analysis, uncertainty estimation, retrieval augmentation, and prompt-level defense. The contribution of the present work is orthogonal to these approaches.

Rather than focusing only on post-generation verification or external knowledge access, we propose that hallucination should also be addressed at the level of **semantic frame construction**. In this sense, the ACE Semantic Gateway is positioned as an architectural layer between raw user input and generative inference. It introduces:

- explicit contextual classification,

- construction of contextual reference fields,
- geometric evaluation through origin energy,
- threshold-based routing before response generation.

This shifts the system from prompt-derived inference toward context-conditioned semantic evaluation, which is the central architectural claim of this paper. The current manuscript already develops this argument through the concepts of Contextual Reference Fields, ACE origin energy, and the implemented semantic gateway prototype.

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3. Background and Problem Formulation

Unlike hallucination detection or retrieval-augmented generation approaches, which operate during or after generation, the ACE Semantic Gateway introduces a deterministic contextual layer prior to inference. This architectural placement allows semantic stabilization before the generative process begins.

3.1 Transformer-Based Language Models

Transformer-based architectures have become the dominant paradigm for modern natural language processing systems. Models such as GPT-style architectures represent tokens as high-dimensional vectors and compute contextual relationships through multi-head attention mechanisms.

During inference, the model processes a sequence of tokens by projecting them into a shared embedding space and iteratively updating their representations through attention layers. The output distribution for the next token is then computed using these contextualized representations.

While this architecture enables powerful pattern recognition and generative capabilities, it also introduces a critical dependency on the **semantic interpretation of the input sequence**. The model must infer the contextual meaning of the prompt in order to generate a continuation that appears coherent within that context.

However, because transformers operate on statistical associations within large embedding spaces, the semantic frame used during generation is not explicitly defined. Instead, it emerges implicitly from the relationships between tokens and previously learned patterns.

This implicit construction of semantic reference frames is a key factor in the instability described in this work.

3.2 Semantic Instability in Language Model Inference

In many practical scenarios, natural language prompts are incomplete, ambiguous, or contextually underspecified. When this occurs, the language model must infer the intended semantic domain probabilistically.

For example, a prompt such as:

“Why is my system failing?”

may refer to multiple domains, including:

- software debugging
- hardware failure
- network configuration
- conceptual systems

Without explicit contextual anchors, the model may construct several candidate interpretations simultaneously. The internal representation space therefore contains multiple semantic directions that may compete during generation.

This phenomenon can lead to responses that are internally coherent but externally misaligned with the intended meaning of the input.

Such behavior is typically described as hallucination.

3.3 Linguistic Entropy

We describe the uncertainty associated with competing semantic interpretations as **linguistic entropy**.

Linguistic entropy arises when:

- contextual signals are insufficient,
- multiple interpretations are equally plausible,
- logical structure is incomplete or ambiguous.

In transformer systems, high linguistic entropy results in attention distributions that are spread across multiple semantic interpretations. As a result, generation may combine elements from incompatible contextual frames.

This diffusion of semantic interpretation increases the probability that the generated response will deviate from the intended meaning of the prompt.

Reducing linguistic entropy therefore requires mechanisms capable of stabilizing the semantic reference frame prior to generation.

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4. ACE Minimum Energy Criterion

4.1 Energy-Based Evaluation of Semantic Alignment

The Axiomatic Criterion Engine (ACE) introduces an energy-based framework for evaluating semantic stability within a contextual reference space.

Given a contextual reference field S, constructed from domain anchors and axiomatic constraints, a candidate input representation z can be evaluated relative to this field.

The ACE origin energy is defined as:

$$O(z) = \| z - PS(z) \|^2$$

where PS(z) denotes the orthogonal projection of the candidate representation onto the contextual subspace S.

This value represents the **semantic deviation of the candidate representation from the contextual reference field**.

4.2 Interpretation of the Origin Energy

The ACE origin energy provides a geometric interpretation of semantic alignment.

Low origin energy indicates that the candidate representation lies close to the contextual subspace, suggesting that the input aligns with the contextual domain.

High origin energy indicates that the representation lies outside the contextual field, suggesting contextual mismatch or semantic instability.

This formulation allows contextual alignment to be evaluated deterministically prior to generation.

4.3 Relationship to Semantic Stability

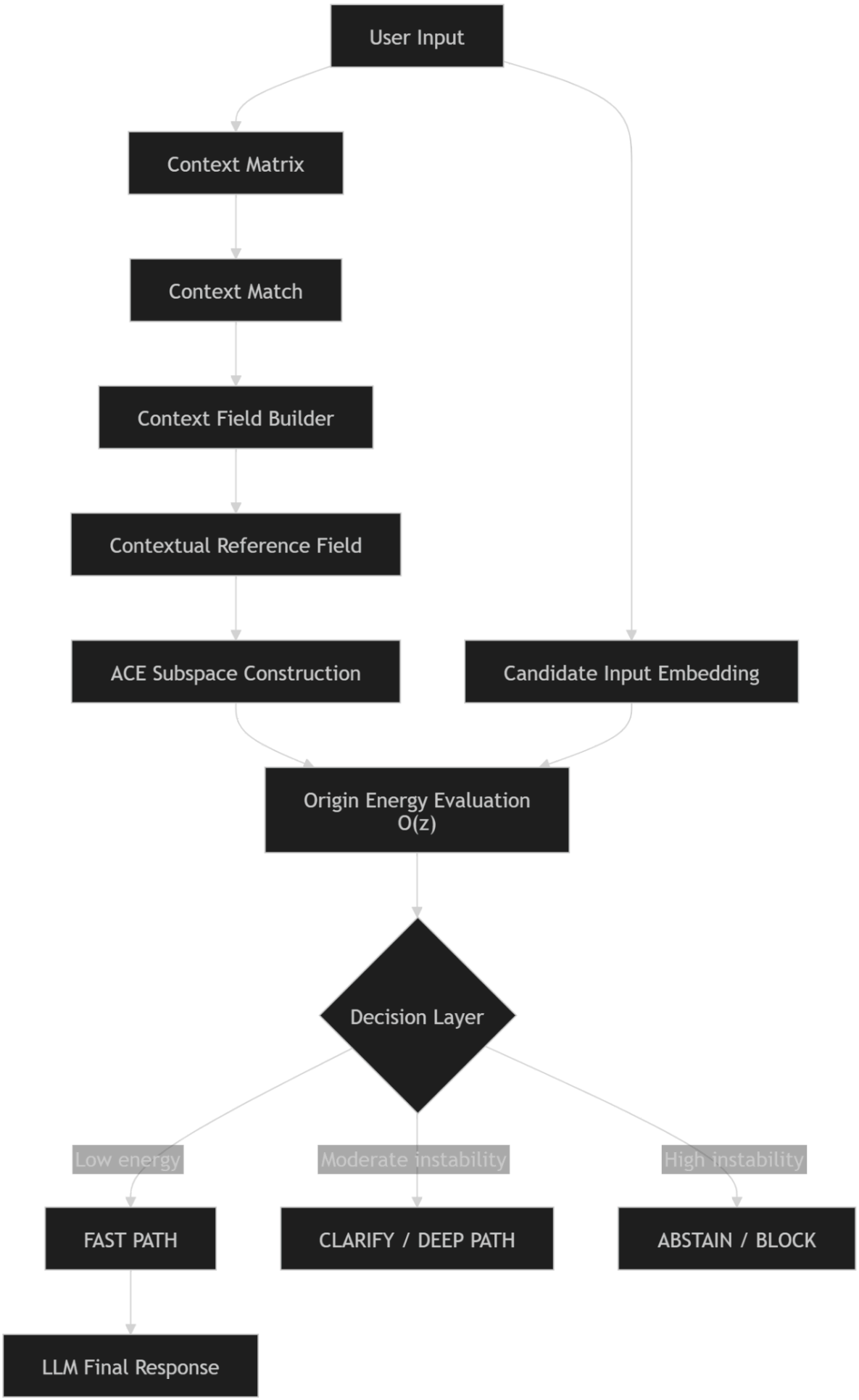
Within the ACE framework, hallucination can be interpreted as generation occurring within an unstable semantic environment.

If the contextual reference field is poorly defined, candidate representations may project onto inconsistent semantic directions within the embedding space.

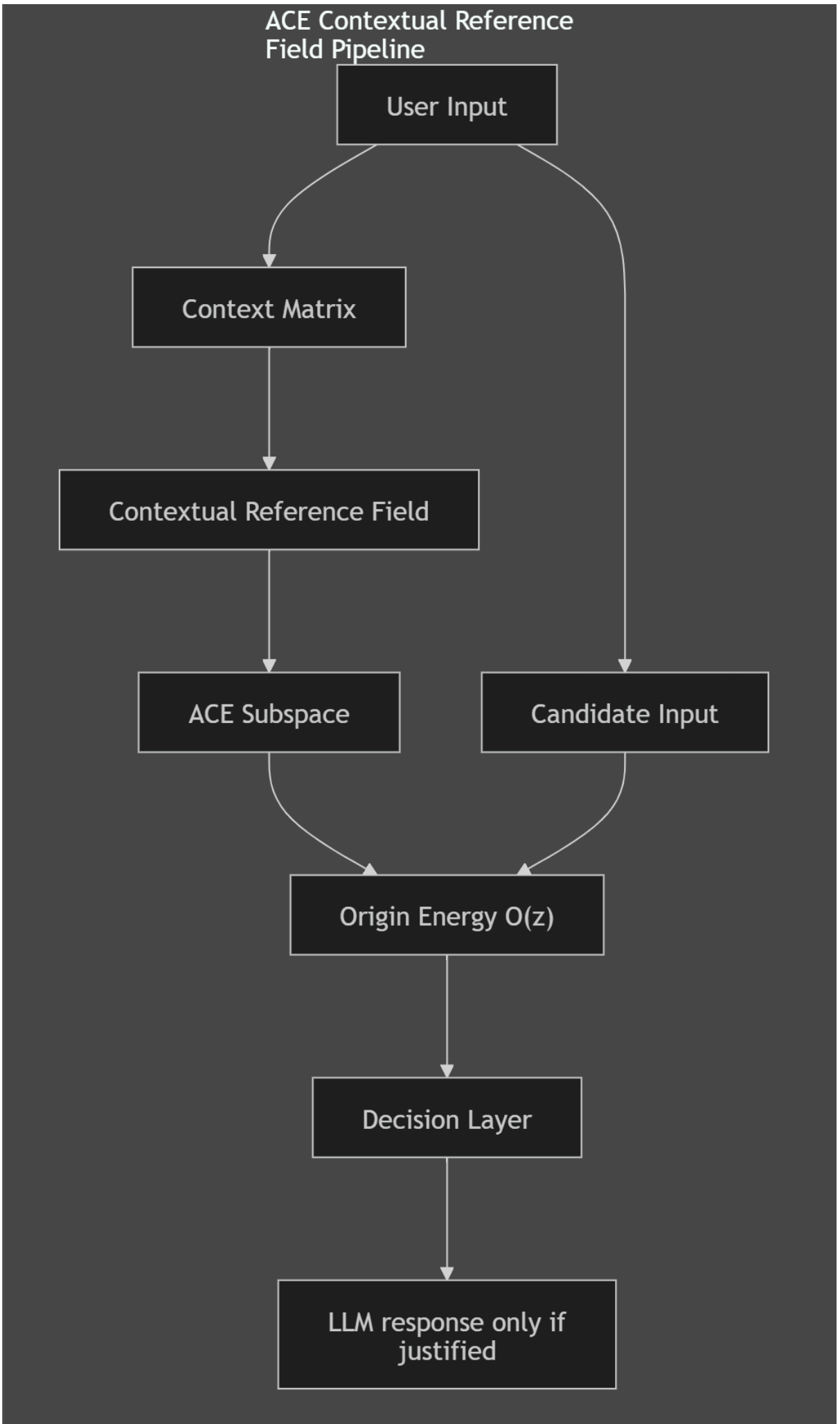
By introducing a structured contextual field and evaluating origin energy relative to that field, the system can detect situations where the semantic frame is unstable.

This enables the inference pipeline to delay generation until contextual alignment is sufficiently established.

Contextual ACE Gateway Architecture

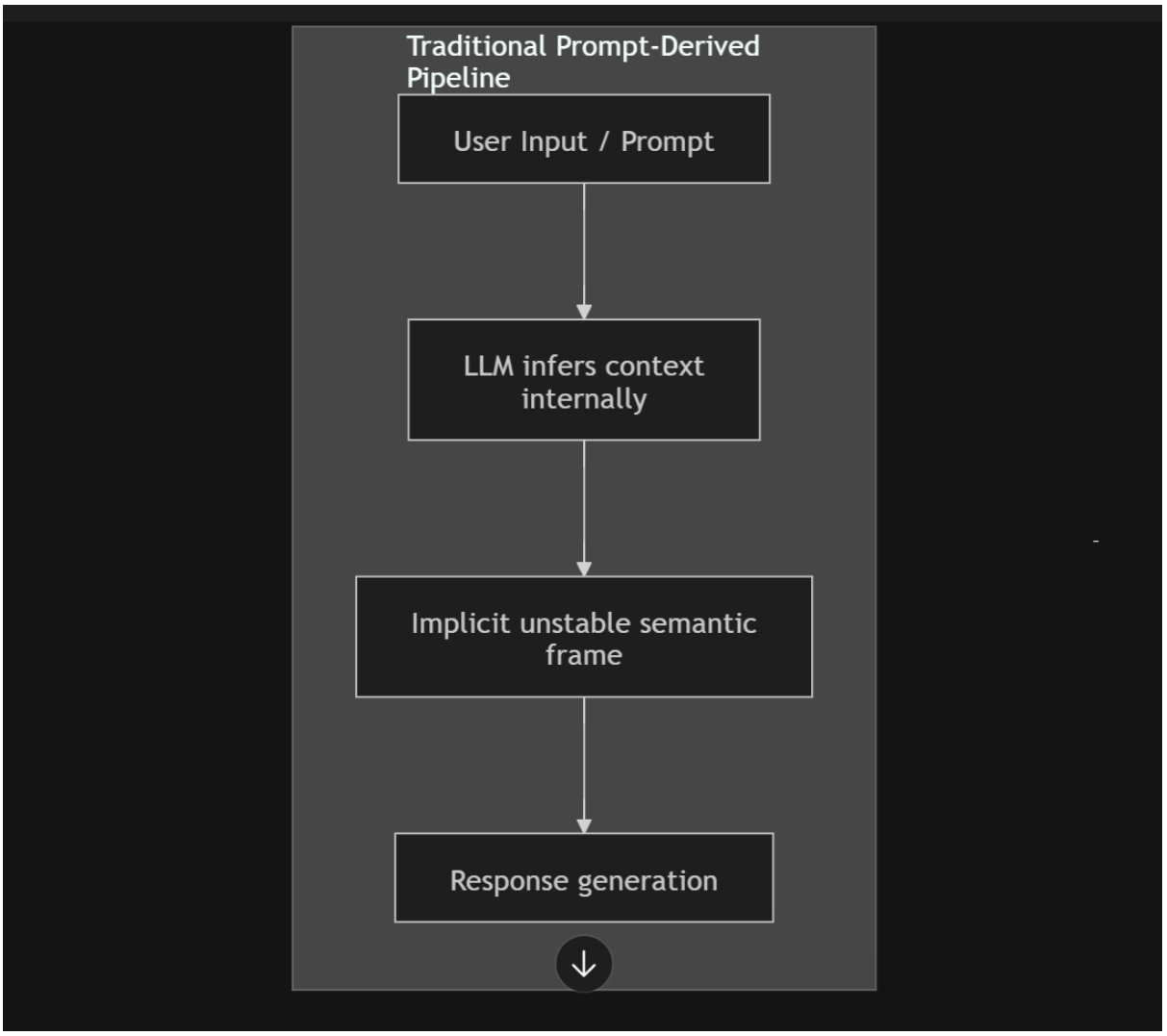


Contextual ACE Gateway Architecture



Contextual ACE Gateway Architecture

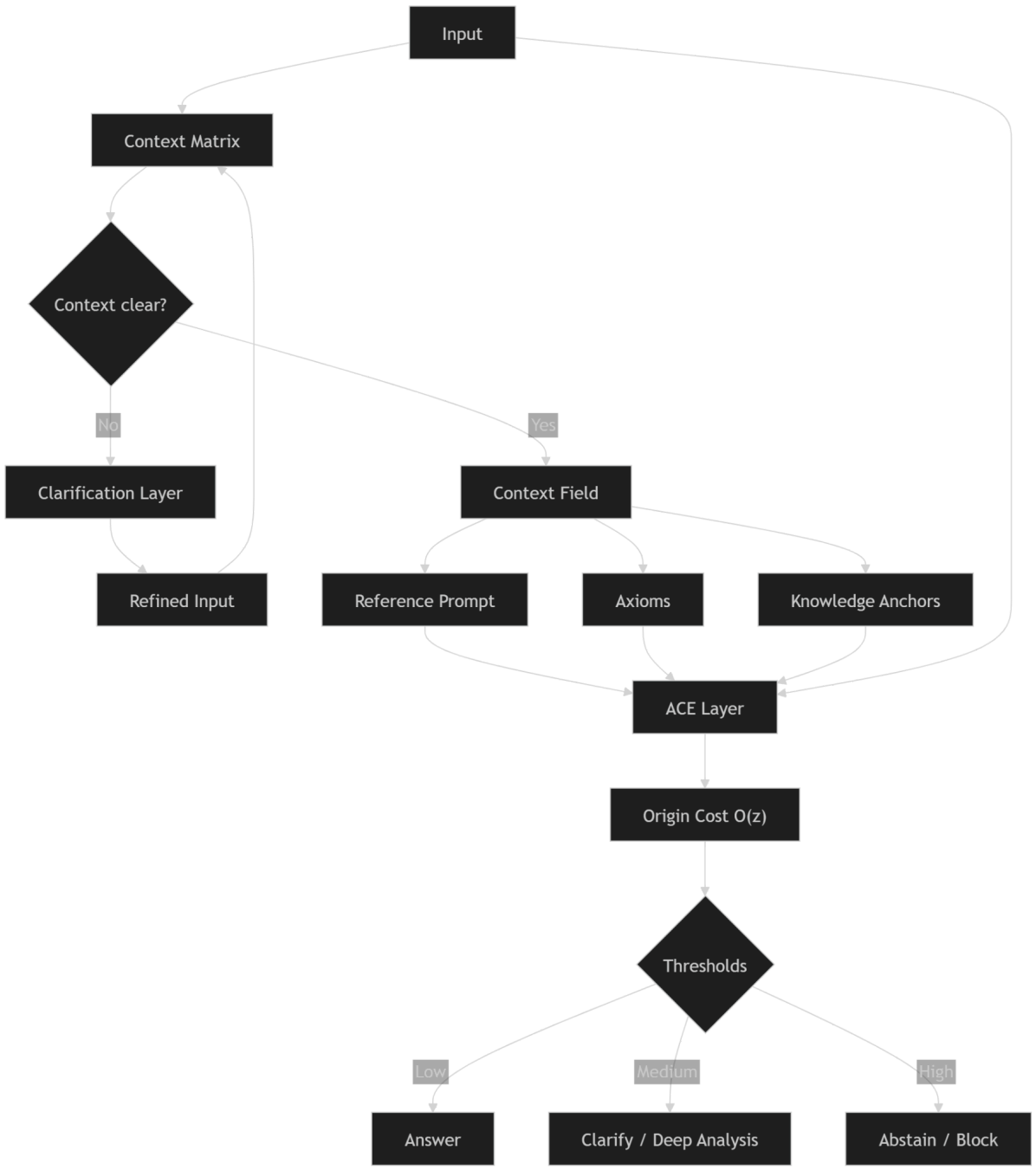
Before prompt → LLM → frame → answer



Now input → context → reference field → ACE → answer

ACE Semantic Gateway Pipeline

Contextual ACE Gateway Architecture



5. Contextual Reference Fields

5.1 The Problem of Prompt-Derived Semantic Frames

Most current large language model (LLM) systems derive their semantic reference frame directly from the input prompt. In practice, this means that the model must simultaneously perform three tasks:

1. infer the intended context of the prompt,
2. construct a semantic space in which the prompt is interpreted,
3. generate a response within that space.

While this process works adequately for many routine tasks, it introduces a fundamental structural weakness. If the prompt itself is ambiguous, incomplete, or internally inconsistent, the semantic frame constructed by the model may be unstable.

In such cases, the model may generate responses that appear coherent but are grounded in an incorrect contextual frame. This phenomenon contributes significantly to what is commonly described as **hallucination in language models**.

The key issue is that the semantic reference space is implicitly derived from the prompt itself. Consequently, the system lacks an external mechanism to verify whether the constructed semantic frame corresponds to a stable interpretation of the request.

5.2 Contextual Reference Fields

To address this limitation, we introduce the concept of a **Contextual Reference Field**.

Instead of deriving the semantic frame directly from the prompt, the system first performs an explicit contextual analysis. The input is evaluated against a structured matrix of contextual domains that represent stable semantic environments.

These contextual domains may include categories such as:

- technical systems
- formal mathematical structures
- psychological experience
- aesthetic interpretation
- ethical reasoning
- spiritual or existential inquiry

Each domain defines a set of contextual anchors including:

- semantic anchors
- axiomatic constraints
- domain-specific knowledge elements

Formally, the contextual matrix can be represented as:

C=[v1,v2,v3,…,vn]

where each vector viv_ivi represents a contextual element within the domain.

The contextual field S is then defined as the span of this matrix:

S=span(C)

This span forms a **stable semantic reference space** within which the prompt can be evaluated.

5.3 Construction of the Contextual Subspace

Once the contextual domain has been identified, the system constructs a reference subspace composed of:

- contextual axioms
- domain knowledge anchors
- semantic reference prompts

These elements collectively define the contextual field against which the input is evaluated.

Let:

- ppp denote the contextual reference prompt,
- A denote the set of axiomatic constraints,
- K denote the set of contextual knowledge anchors.

The reference subspace is constructed from the embeddings of these elements.

The candidate input z is then evaluated relative to this subspace using the **ACE origin energy criterion**.

5.4 Construction of the Contextual Matrix

The contextual matrix used by the ACE Semantic Gateway is constructed from a set of domain profiles representing coherent regions of semantic alignment. Each profile contains structured contextual elements that help determine whether an input query belongs to a given semantic domain.

These contextual elements include:

- **lexical anchors**, which represent domain-specific terminology,
- **semantic anchors**, which represent broader conceptual associations,
- **intent markers**, which indicate the type of reasoning requested by the query,
- **negative markers**, which help reduce false contextual matches.

Together, these elements define a contextual profile that can be embedded into vector space and compared against the incoming query. The collection of such profiles forms the **contextual matrix**, where each profile corresponds to a region of semantic alignment used to construct a **contextual reference field**.

During inference, the query is evaluated against this matrix to determine the most coherent contextual region before generation occurs.

5.5 Origin Energy within a Contextual Field

The ACE framework evaluates the stability of a candidate representation relative to a reference subspace.

Given the contextual field S, the origin energy of a candidate representation z is defined as:

$O(z) = \| z - PS(z) \|_2$

where:

- PS(z) denotes the orthogonal projection of z onto the contextual subspace S.

Intuitively, this value measures the **semantic deviation of the candidate input from the contextual field**.

Low origin energy indicates that the input aligns with the contextual domain, while high energy indicates semantic instability or contextual mismatch.

5.6 Contextual ACE Gateway Architecture

Figure 1 illustrates the architecture of the **Contextual ACE Gateway**.

The key architectural shift introduced by this framework is that the semantic reference space is no longer derived implicitly from the prompt. Instead, it is constructed explicitly through contextual analysis prior to generation.

The system operates according to the following sequence:

1. **Context Detection**
   The input is evaluated against a contextual matrix to determine the most appropriate semantic domain.
2. **Context Field Construction**
   A structured reference field is constructed from contextual axioms, knowledge anchors, and semantic descriptors.
3. **ACE Subspace Formation**
   The contextual field is embedded to construct a reference subspace.
4. **Origin Energy Evaluation**
   The candidate input is evaluated relative to this subspace using the ACE origin energy criterion.

5. **Decision Layer**
   Based on the resulting energy value, the system determines whether to:
   - generate a response,
   - request clarification,
   - abstain from answering.

This architecture separates **context determination** from **response generation**, thereby reducing the risk of semantic instability arising from prompt-derived reference frames.

ContextMatrix
↓
ContextField
↓
ACE Layer
↓
Origin Cost
↓
Gateway Decision

6. Why Prompt-Derived Subspaces Produce Hallucination

6.1 Competing Interpretations in Transformer Systems

Transformer-based language models process text by projecting tokens into high-dimensional embedding spaces and computing contextual relationships through attention mechanisms. During inference, the model must interpret the input prompt within this semantic space and generate a continuation consistent with the inferred context.

However, natural language often contains implicit assumptions, incomplete references, and multiple potential interpretations. When a prompt lacks sufficient contextual grounding, the model may construct several competing semantic frames simultaneously.

For example, a question such as:

> *Why is my system failing?*

could plausibly refer to:

- a software configuration issue,
- a hardware malfunction,
- a network connectivity problem,
- a conceptual or theoretical system.

Without a clear contextual anchor, the model must infer the intended semantic frame probabilistically. In such cases, multiple plausible interpretations coexist within the model's embedding space.

This situation can be described as **semantic competition**, where several candidate interpretations compete for dominance during generation.

6.2 Linguistic Entropy and Contextual Instability

We refer to the degree of uncertainty in the semantic interpretation of a prompt as **linguistic entropy**.

Linguistic entropy arises when:

- key contextual information is missing,
- multiple interpretations are equally plausible,
- the logical structure of the prompt is ambiguous.

In high-entropy prompts, the model's internal attention mechanisms distribute probability mass across several potential interpretations. The resulting attention distribution becomes diffuse, which increases the likelihood that the generated response will incorporate elements from incompatible contextual frames.

This process can produce outputs that appear coherent at the sentence level but are not grounded in a stable semantic interpretation of the input.

Such responses are typically described as **hallucinations**, though the underlying cause is not random error but **contextual instability in the semantic reference frame**.

6.3 Prompt-Derived Subspaces

In most current LLM architectures, the semantic reference space used during generation is implicitly derived from the prompt itself.

Formally, if the prompt representation is denoted by vector ppp, the model constructs an implicit semantic neighborhood around this representation and performs generation within that region of the embedding space.

However, when the prompt is ambiguous, the resulting semantic neighborhood may include vectors corresponding to incompatible contextual interpretations.

This produces an **unstable semantic subspace**, within which projections of candidate responses may appear locally coherent while remaining globally inconsistent with the intended meaning.

Consequently, projecting candidate responses within a prompt-derived subspace does not guarantee semantic validity.

This observation highlights a structural limitation of prompt-derived inference: the model cannot reliably determine whether the semantic reference space itself is valid.

6.4 Contextual Reference Fields as Stability Anchors

The contextual architecture introduced in this work addresses this limitation by separating **context identification** from **response generation**.

Instead of deriving the semantic frame directly from the prompt, the system first identifies the contextual domain in which the input is most plausibly interpreted.

This contextual domain defines a **reference field** consisting of:

- semantic anchors,
- axiomatic constraints,
- domain-specific knowledge elements.

These elements form a structured semantic environment from which a stable reference subspace can be constructed.

By embedding these contextual elements and computing their span, the system constructs a semantic reference space that is independent of the prompt's ambiguity.

The candidate input is then evaluated relative to this contextual field.

6.5 Energy-Based Evaluation of Contextual Alignment

Within the ACE framework, semantic stability is evaluated using an origin energy function.

Let SSS denote the contextual reference subspace constructed from contextual anchors and domain knowledge.

Given a candidate representation z, the origin energy is defined as:

$O(z) = // z - PS(z) // 2$

where PS(z) represents the orthogonal projection of z onto the contextual subspace S.

This quantity measures the degree to which the candidate representation deviates from the contextual field.

If the candidate aligns with the contextual domain, the projection will lie close to the candidate representation and the origin energy will remain low.

Conversely, if the candidate representation lies outside the contextual field, the residual distance will increase, indicating semantic instability.

This energy-based formulation provides a deterministic criterion for evaluating contextual alignment prior to generation.

6.6 Architectural Implications

The introduction of contextual reference fields changes the architecture of language model inference in a fundamental way.

Instead of relying on the prompt to implicitly define the semantic frame, the system performs the following steps:

1. contextual domain identification,
2. construction of a contextual reference field,
3. formation of a reference subspace,
4. energy-based evaluation of candidate inputs,
5. generation only when contextual alignment is satisfied.

By externalizing the construction of the semantic reference space, the system reduces the risk of generation within unstable prompt-derived subspaces.

In practical implementations, this architecture can function as a **semantic gateway layer** placed before the language model. The gateway evaluates contextual stability and determines whether the input should be answered directly, clarified, or rejected.

This design provides a deterministic mechanism for mitigating hallucination while preserving the generative capabilities of modern transformer-based models.

ContextMatrix
↓
ContextField
↓
ACE Origin Energy
↓
Gateway Decision

7. Experimental Demonstration and Practical Implementation

7.1 Implementation Overview

To evaluate the feasibility of contextual reference fields in practical systems, we implemented a prototype middleware architecture referred to as the **ACE Semantic Gateway**.

The purpose of the gateway is to act as a semantic evaluation layer placed before the language model. Instead of allowing the model to infer the semantic frame directly from the prompt, the gateway performs contextual analysis and energy-based alignment prior to generation.

The implementation consists of four primary components:

1. **Context Matrix**
   A structured matrix of contextual domains used to determine the semantic environment of the input.
2. **Context Field Builder**
   A module that constructs a contextual reference field from domain-specific anchors, axioms, and knowledge descriptors.
3. **ACE Evaluation Layer**
   A mathematical layer that evaluates the alignment of the input representation with the contextual reference subspace using the origin energy criterion.
4. **Decision Gateway**
   A routing mechanism that determines whether the input should be answered directly, clarified, or rejected based on energy thresholds.

Together, these components form a deterministic semantic gateway that operates independently of the language model.

7.2 System Pipeline

The operational pipeline of the ACE Semantic Gateway is illustrated in Figure 2.

The system processes an input request according to the following sequence:

1. **Input Reception**
   The user input is received by the gateway.
2. **Context Detection**
   The input is evaluated against a contextual matrix containing predefined semantic domains.

3. **Contextual Field Construction**
   Once a contextual domain is identified, the system constructs a reference field composed of:
   - contextual axioms,
   - semantic anchors,
   - domain knowledge elements.
4. **Reference Subspace Formation**
   These elements are embedded and used to construct a contextual subspace.
5. **Origin Energy Evaluation**
   The candidate input representation is evaluated relative to this contextual subspace.
6. **Decision Routing**
   Based on the resulting energy value, the gateway selects one of three possible outcomes:
   - direct response,
   - contextual clarification,
   - abstention.

This architecture separates contextual interpretation from response generation, thereby reducing the probability of unstable semantic inference.

7.3 Prototype Implementation

A prototype implementation of the ACE Semantic Gateway was developed in Python as a modular middleware layer.

The system includes the following modules:

ContextMatrix
ContextFieldBuilder
ACELayer
SemanticGateway

The **ContextMatrix** module performs contextual classification of the input using a combination of lexical signals, semantic anchors, and embedding similarity.

The **ContextFieldBuilder** constructs a contextual reference field containing:

- domain axioms,
- contextual knowledge elements,
- semantic anchors.

The **ACELayer** computes the origin energy of the input representation relative to the contextual subspace defined by these elements.

Finally, the **SemanticGateway** integrates these components and applies threshold-based decision logic.

This modular design allows the gateway to operate independently of any specific language model implementation.

7.4 Experimental Observations

Preliminary experiments were conducted using a set of representative prompts spanning multiple contextual domains.

Example input categories included:

- existential questions,
- technical troubleshooting,
- mathematical reasoning,
- psychological inquiry,
- aesthetic interpretation,
- ethical dilemmas.

For each input, the system performed contextual classification and evaluated semantic alignment using the ACE origin energy criterion.

Example outputs include:

Input	Context	Origin Energy
Meaning of life	Spiritual / Existential	Low
Server boot failure	Technical Systems	Low
Derivative proof	Formal Mathematics	Moderate
Human anxiety	Psychology	Low
Painting aesthetics	Aesthetic Interpretation	Low
Moral truth dilemma	Ethics	Low

Across these examples, the gateway consistently identified the appropriate contextual domain and produced low origin energy values for inputs that aligned with the contextual reference field.

This behavior suggests that contextual reference fields provide a stable semantic environment for evaluating input alignment prior to generation.

7.5 Interpretation of Results

The experimental observations support the hypothesis that contextual reference fields reduce semantic instability during language model inference.

When the contextual domain is explicitly identified and used to construct the reference subspace, the origin energy of the input representation remains low, indicating strong alignment with the contextual field.

Conversely, when inputs deviate from the contextual domain, the origin energy increases, signaling potential semantic instability.

This property enables the gateway to detect situations in which generation should be delayed or clarified.

Importantly, this mechanism does not rely on probabilistic heuristics or post-generation filtering. Instead, it evaluates contextual alignment prior to generation using a deterministic geometric criterion.

7.6 Practical Implications

The ACE Semantic Gateway architecture provides several practical advantages for real-world AI systems.

First, it reduces the likelihood of hallucination by preventing generation within unstable semantic frames.

Second, it introduces a deterministic evaluation layer that can operate independently of the language model's internal architecture.

Third, the contextual matrix can be extended incrementally, allowing systems to incorporate domain knowledge without retraining the underlying model.

Finally, the architecture enables a more interpretable inference process. Because contextual classification and origin energy evaluation occur explicitly, system behavior becomes easier to analyze and debug.

These properties make the contextual gateway approach particularly suitable for applications requiring high reliability, such as scientific reasoning, technical assistance, and safety-critical decision systems.

ContextMatrix

↓

ContextField

↓

ACELayer

Contextual ACE Gateway Architecture

↓

SemanticGateway

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8. Limitations and Future Work

While the Contextual ACE Gateway architecture demonstrates promising properties, several limitations remain.

First, the contextual matrix used in the current implementation is defined manually using predefined contextual domains and semantic anchors. Although this structure provides strong interpretability, future work may explore mechanisms for dynamically extending the contextual matrix through automated knowledge extraction.

Second, the prototype implementation relies primarily on lexical signals and simplified embedding similarity for contextual classification. Integrating high-quality embedding models may improve the robustness of contextual detection, particularly in complex or highly technical domains.

Third, the current implementation evaluates contextual alignment using static contextual fields. Future work could investigate adaptive contextual fields that evolve based on interaction history or domain-specific knowledge acquisition.

Another promising direction involves combining contextual reference fields with retrieval-based systems. In such architectures, retrieved knowledge could contribute directly to the construction of contextual fields, enabling more precise domain anchoring.

Finally, large-scale empirical evaluation across diverse language tasks will be necessary to quantify the impact of contextual reference fields on hallucination reduction and semantic stability in production environments.

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9. Conclusion

This work introduces an alternative architectural approach to language model inference based on contextual reference fields.

Current large language model systems typically derive their semantic reference frame directly from the input prompt. While this approach enables flexible generative behavior, it also introduces the possibility of semantic instability when prompts are ambiguous or contextually underspecified.

Contextual ACE Gateway Architecture

We argue that hallucinations in language models are partly a consequence of this architectural property.

To address this limitation, we propose a framework that separates **context identification** from **response generation**. The system first determines the contextual domain of the input using a structured contextual matrix, then constructs a contextual reference field composed of domain anchors, axioms, and knowledge elements.

Candidate inputs are evaluated relative to this field using the ACE origin energy criterion, which measures semantic deviation from the contextual subspace.

We implement this approach in a prototype middleware architecture, the **ACE Semantic Gateway**, which performs contextual classification, constructs contextual reference fields, and evaluates semantic alignment prior to generation.

The resulting architecture transforms the inference pipeline from a prompt-derived process

prompt → LLM → semantic frame → answer

into a context-anchored process

input → context → reference field → ACE → answer

By externalizing the construction of the semantic reference frame, this approach provides a deterministic mechanism for reducing semantic instability while preserving the generative capabilities of modern transformer-based models.

The central claim of this work is that hallucination is not only a problem of factual grounding or decoding strategies.
It is fundamentally a problem of **semantic frame instability**.

By introducing contextual reference fields prior to inference, the ACE Semantic Gateway reframes language model interaction from prompt-driven exploration toward **context-conditioned semantic convergence**.

Hallucination is not only a failure of factual grounding but a consequence of semantic frame instability.

Contextual reference fields provide a deterministic mechanism for stabilizing that frame before generation begins.

Final Pipeline

user query
↓
context_matrix
↓
context classification
↓

Contextual ACE Gateway Architecture

ACE origin energy evaluation
↓
LLM generation

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10 Limitations / Future Work:

In large-scale deployments the contextual matrix can be extended dynamically using embedding clustering, domain ontologies, or learned contextual profiles. The present work demonstrates the deterministic framework rather than a fully automated contextual discovery pipeline.

The current implementation demonstrates the deterministic contextual architecture. Extending the contextual matrix through automated discovery or learned contextual profiles remains an important direction for future work.

Contribution Statement

This paper introduces the concept of contextual reference fields as a deterministic semantic stabilization layer prior to LLM inference.

ACE Mathematical Appendix (LaTeX)

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\section*{Mathematical Appendix: ACE Minimum Energy Criterion}

\subsection*{Purpose}

This appendix provides a mathematical interpretation of the
\textbf{ACE Minimum Energy Criterion} and its role in stabilizing semantic
inference in large language models.

The goal is to formalize the geometric intuition behind:

\begin{itemize}
\item contextual reference fields
\item semantic attractors
\item energy minimization
\item linguistic entropy reduction
\end{itemize}

These notes describe the mathematical structure underlying the architecture
presented in the ACE Semantic Gateway.

\subsection*{1. Embedding Space Representation}
```

Contextual ACE Gateway Architecture

Let

$$
E \subset \mathbb{R}^d
$$

be the embedding space used by a language model.

Each token or sequence is represented as a vector

$$
z \in E
$$

Semantic similarity between expressions is commonly approximated using cosine similarity:

$$
\text{sim}(x,y) =
\frac{x \cdot y}{\|x\| \|y\|}
$$

Although embedding spaces learned by transformers are not perfectly linear semantic manifolds, they exhibit locally structured geometry that enables meaningful vector operations.

\subsection*{2. Contextual Anchor Matrix}

A contextual domain is represented by a set of anchor vectors

$$
C = [v_1, v_2, \dots, v_n]
$$

where

$$
v_i \in E
$$

These anchors may correspond to:

\begin{itemize}
\item domain concepts
\item axiomatic principles
\item canonical statements
\item contextual prompts
\end{itemize}

The anchors define the semantic structure of the contextual domain.

\subsection*{3. Contextual Reference Subspace}

The contextual reference field is defined as the span of the anchor vectors:

$$
S = \text{span}(C)
$$

Thus

$$
S \subset E
$$

represents a linear subspace corresponding to a contextual semantic domain.

Any vector within this subspace can be written as

$$
z = \sum_{i=1}^{n} \alpha_i v_i
$$

for some coefficients α_i.

\subsection*{4. Orthogonal Projection}

Given a candidate semantic vector

$$
z \in E
$$

its orthogonal projection onto the contextual subspace is

$$
P_S(z)
$$

If the anchor matrix C has full column rank, the projection operator can be expressed as

$$
P_S = C(C^T C)^{-1}C^T
$$

Thus

$$
P_S(z) = C(C^T C)^{-1}C^T z
$$

This projection extracts the semantic component aligned with the contextual domain.

5. ACE Origin Energy

The ACE Minimum Energy Criterion measures the deviation of a candidate representation from the contextual subspace.

The energy function is defined as

$$
O(z) = \|z - P_S(z)\|^2
$$

where

- z is the candidate semantic vector
- $P_S(z)$ is the contextual projection

Interpretation:

- $O(z) = 0$ indicates perfect contextual alignment
- small $O(z)$ indicates strong semantic consistency
- large $O(z)$ indicates contextual deviation

ACE therefore prefers candidate responses that minimize this energy.

6. Energy Minimization

Semantic inference can be interpreted as an energy minimization process.

Given contextual field S, the objective is

$$
z^* = \arg\min_z O(z)
$$

subject to the constraints imposed by the language model.

This formulation introduces a geometric constraint on the inference process.

\subsection*{7. Linguistic Entropy}

Let a prompt p admit several possible contextual domains

$$
S_1, S_2, \dots, S_k
$$

Let

$$
P(S_i | p)
$$

denote the probability that prompt p belongs to domain S_i.

The linguistic entropy of the prompt can be defined as

$$
H(p) =
- \sum_{i=1}^{k} P(S_i | p) \log P(S_i | p)
$$

High linguistic entropy indicates that the prompt is compatible with multiple semantic interpretations.

Such prompts are more likely to produce unstable inference.

\subsection*{8. Context Selection as Entropy Reduction}

The ACE architecture reduces linguistic entropy by selecting a contextual domain before generation:

$$
p \rightarrow S^* = \arg\max_i P(S_i | p)
$$

Once the contextual field is selected, inference proceeds within that semantic region.

\subsection*{9. Semantic Attractors}

Embedding space can be interpreted as containing regions of semantic stability corresponding to conceptual domains.

These regions behave as \textbf{semantic attractors}.

Inference trajectories tend to converge toward these attractors.

Ambiguous prompts may lie near boundaries between attractors.

ACE stabilizes inference by explicitly selecting the contextual attractor before generation begins.

\subsection*{10. Energy Landscape Interpretation}

The embedding space can be interpreted as an energy landscape in which contextual reference fields correspond to stable valleys.

The ACE criterion evaluates candidate responses relative to these valleys.

Responses outside the contextual region correspond to higher energy states.

\subsection*{11. Relation to Transformer Inference}

Transformer models generate tokens using attention-based updates of contextual embeddings.

However, transformers do not explicitly enforce a global semantic reference frame.

Instead, the semantic frame emerges implicitly from token relationships.

ACE introduces an explicit geometric constraint that stabilizes the semantic frame prior to generation.

\subsection*{12. Approximate Implementations}

In practical systems, exact projection may be approximated through:

\begin{itemize}
\item cosine similarity with contextual anchors
\item embedding clustering
\item heuristic contextual scoring
\item simplified energy thresholds
\end{itemize}

These approximations preserve the geometric intuition while enabling efficient implementation.

\subsection*{13. Toward a Theory of Semantic Stability}

The ACE framework suggests a broader research direction: a mathematical theory

of semantic stability in embedding spaces.

Potential areas of investigation include:

\begin{itemize}
\item stability of contextual attractors
\item entropy reduction through contextual constraints
\item geometric interpretation of hallucination
\item dynamical models of semantic inference
\end{itemize}

\subsection*{Conclusion}

The ACE Minimum Energy Criterion interprets semantic alignment as a geometric constraint in embedding space.

By introducing contextual reference fields and minimizing semantic energy relative to those fields, inference is stabilized before generation occurs.

This perspective reframes hallucination as a problem of \textbf{semantic instability} rather than purely probabilistic error.